

Lecture 14



Coherence

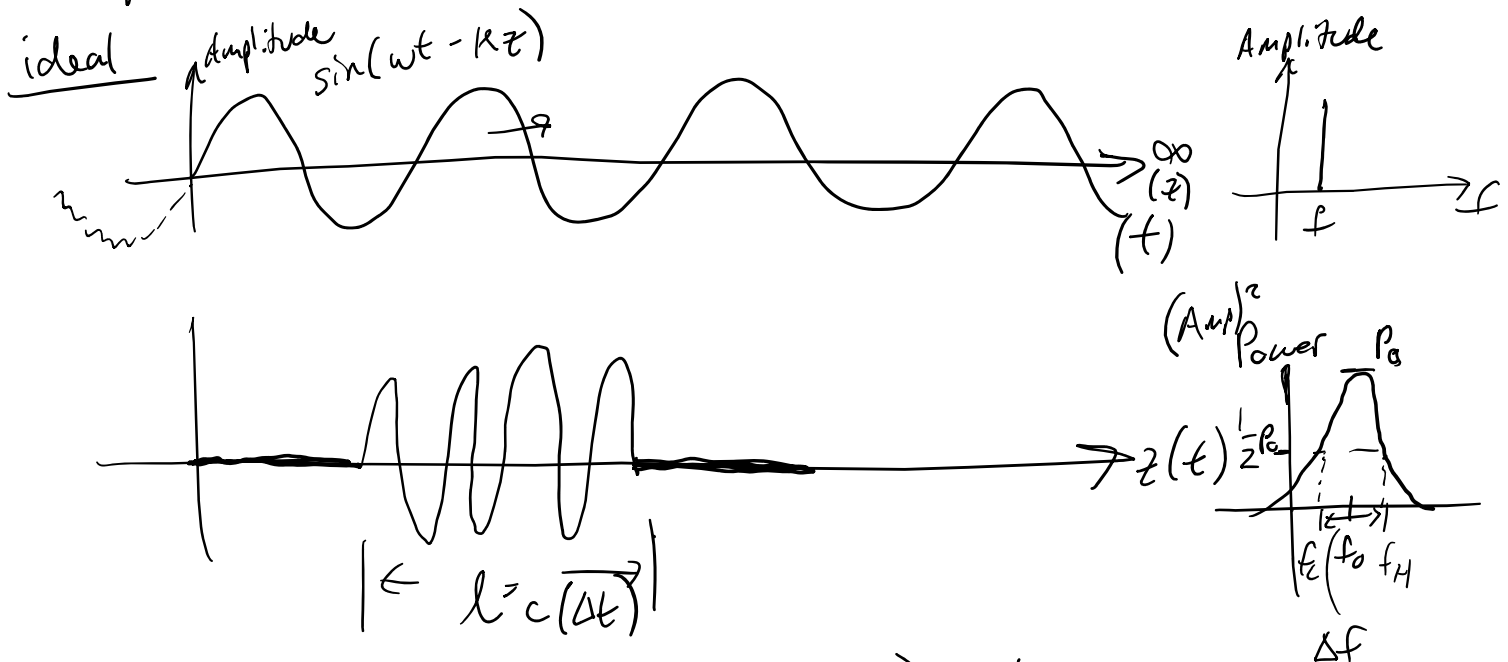


Incoherence



Partial Coherence

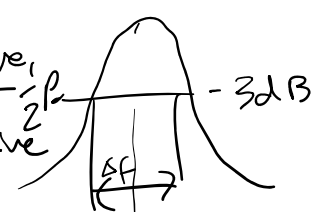
* coherence (optical) means we can predict the phase of a wave and any portion or parameter of that wave anywhere over space and time.



Full width at half maximum (FWHM) = $\frac{1}{2} P_{\text{peak}}$

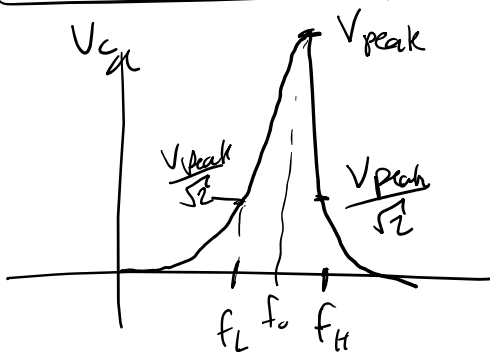
$$\frac{|\vec{E}_{\text{peak}}|}{\sqrt{2}} \cdot \frac{H_{\text{peak}}}{\sqrt{2}} = \frac{P_{\text{peak}}}{2}$$

If given an E-field; $\text{FWHM} \rightarrow \frac{E_0}{\sqrt{2}}$

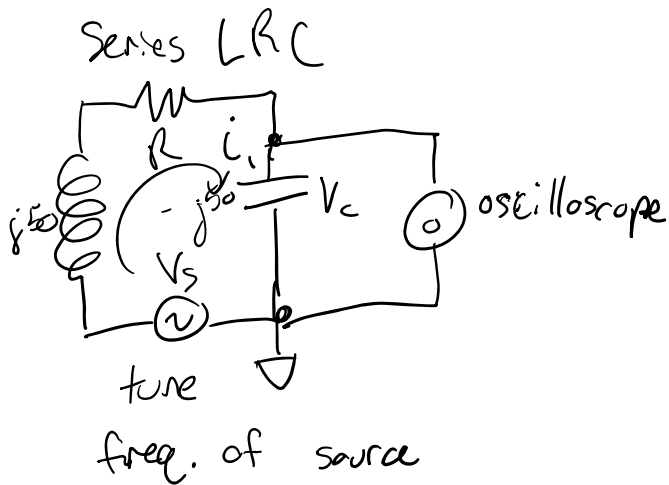
$$\text{Quality factor} = \frac{P_{\text{stored}}}{P_{\text{loss}}} = \frac{\text{Active } \frac{1}{2}P}{\text{Reactive } \frac{1}{2}P}$$


$$P_{\text{dB}} = 10(\log_{10}(\frac{1}{2})) = -3\text{dB}$$

In circuit analogy



$$Q = \frac{f_0}{\Delta f} = \frac{f_0}{f_H - f_L}$$



at resonance $X_L = j50\Omega$
 $X_C = -j50\Omega$

$$-V_S + j50i + iR + -j50i = 0$$

$$R = 0.1\Omega$$

$$V_S = (0.1 + j(50 - 50))i \quad V = 100\text{V}$$

$$V_S = 0.1i$$

$$i = \frac{100}{0.1} = 1000\text{A}$$

$$P = 0.1(1000)^2$$

$$P = 100\text{kW}$$

$$\text{VARs} = jX_L(1000)^2 = 50(1000)^2 = 50\text{MVARs}$$

$$Q = \frac{50\text{M}}{100\text{k}} = 500$$

optics

$$\boxed{I \rightarrow II = \frac{1}{2}(\sqrt{\epsilon_r \epsilon_0}) E_0^2}$$

↑
irradiance

↑
intensity

$$\sqrt{\epsilon_r \epsilon_0} = \frac{c}{v}$$

$$\tan^{-1}\left(\frac{x}{R}\right) = \left(\frac{90}{0.1}\right) \rightarrow \sim \cancel{90}^\circ 89.4^\circ$$

$$pf = \cos \theta =$$